Computing Colimits in Higher Categories

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2/27/2017

In the last talk ([Ba17]) we focused on the definition of a colimit in the context of higher categories. We noticed that although the definition is quite satisfying, it is often not actually computable. Thus we will certain techniques that actually allow us to compute some colimits in a higher category. This is the focus of todays talk.

Before we start recall the following definition.

Definition 0.1. For a simplicial map $p\colon S \to \mathcal{C}$, a map $\widetilde{p}\colon S^{\triangleright} \to \mathcal{C}$ extending p is a *colimit* of p if the restriction map

$$\mathcal{C}_{\widetilde{p}/} \to \mathcal{C}_{p/}$$

is a trivial fibration.

Cofinal maps

Cofinal maps allow us to simplify colimit computations by replacing one complicated diagram by a simpler one which has the same result. Most of the material here follows the work of Lurie in Section 4.1 ([Lu09]).

Last time, we used right anodyne maps as an analogue to cofinal inclusions, with the following fact.

Fact 1.1. If $j\colon T \to S$ is right anodyne, then for any $p\colon S \to \mathcal{C}$, the restriction map

$$\mathcal{C}_{p/} \to \mathcal{C}_{pj/}$$

is a trivial fibration.

However, right anodyne maps are very nice maps. We only really used the fact that this map is a categorical equivalence, which would hold even if j is itself any categorical equivalence. So, it makes sense to introduce the following more general definition.

Definition 1.2. A map $f\colon T \to S$ is *cofinal* if for every quasicategory $\mathcal{C}$ and every map $p\colon S \to \mathcal{C}$, the induced map

$$\mathcal{C}_{p/} \to \mathcal{C}_{pf/}$$

is a categorical equivalence.

This is as fine of a definition as any, but what's really good is that we can characterize cofinal maps in other ways. Namely, recall that for a simplicial set $\mathcal{C}$, we have the "contravariant model structure" on $\mathrm{sSet}_{/\mathcal{C}}$ modeling right fibrations. In particular, the contravariant equivalences are maps

$$\begin{tikzcd} T \arrow[rr, "f"] \arrow[dr] & & S \arrow[dl, "p"] \\ & \mathcal{C} & \end{tikzcd}$$

such that

$$T^{\triangleright} \cup_T \mathcal{C} \to S^{\triangleright} \cup_S \mathcal{C}$$

is a categorical equivalence. These contravariant equivalences can be used to help characterize cofinal maps, and as a result cofinal maps can be used to help understand contravariant equivalences. Namely, what's true is the following.

Fact 1.3. A map $f\colon T \to S$ is *cofinal* if for any map $p\colon S \to \mathcal{C}$, the map

$$\begin{tikzcd} T \arrow[rr, "f"] \arrow[dr] & & S \arrow[dl, "p"] \\ & \mathcal{C} & \end{tikzcd}$$

is a contravariant equivalence in $\mathrm{sSet}_{/\mathcal{C}}$.

In other words, cofinal maps are those maps which are always contravariant equivalences.

Remark 1.4. It is actually enough to check that

$$\begin{tikzcd} T \arrow[rr, "f"] \arrow[dr] & & S \arrow[dl, equal] \\ & S & \end{tikzcd}$$

is a contravariant equivalence in $\mathrm{sSet}_{/S}$.

It is also worth mentioning the following facts.

Fact 1.5.

(i) A map j is cofinal and monic if and only if j is right anodyne.

(ii) A map q is cofinal and a right fibration if and only if q is a trivial fibration.

(iii) A map f is cofinal if and only if $f = qj$ with j right anodyne and q a trivial fibration.

One surprising thing that comes out of considering cofinal maps is the following.

Fact 1.6. Let S be any simplicial set. Then there is a cofinal map $NP \to S$ where P is an ordinary category. In fact, P may be taken to be a poset, and this map can be made functorial in S.

So, in generalizing colimits in ordinary categories to colimits in quasicategories, there is no extra generality obtained in considering higher dimensional diagram shapes.

Building colimits out of smaller colimits

The previous fact shows that a quasicategory $\mathcal{C}$ is cocomplete if and only if $\mathcal{C}$ admits colimits of diagrams in the shape of a poset. Actually, we can do much better than this. First, recall the following classical fact.

Proposition 2.1. Let $\mathcal{C}$ be an ordinary category. Then $\mathcal{C}$ is cocomplete if and only if $\mathcal{C}$ has coproducts and coequalizers.

Proof. For any functor $F\colon I \to \mathcal{C}$, we have

$$\operatorname{colim} F = \operatorname{Cok}\left[\coprod_{f\colon i \to i'} Fi \rightrightarrows \coprod_{i''} Fi''\right].$$

□

As one would expect, the corresponding statement is true for quasicategories.

Fact 2.2. Let $\mathcal{C}$ be a quasicategory. Then $\mathcal{C}$ is cocomplete if and only if $\mathcal{C}$ has coproducts and coequalizers.

As an example, we can prove the following. The reader is advised to find his own proof rather than try to follow this one.

Proposition 2.3. Let $\mathcal{C}$ be a quasicategory. If $\mathcal{C}$ admits coproducts and coequalizers, then $\mathcal{C}$ admits pushouts.

Proof. Here, a coequalizer should be taken to be the colimit of a map from the simplicial set

$$\begin{tikzcd} 0 \arrow[r, bend left] \arrow[r, bend right] & 3 \end{tikzcd}.$$

In our proof, we will rather want to take the colimit of a map from the simplicial set

$$\begin{tikzcd} 0 \arrow[r] \arrow[d] & 1 \arrow[d] \\ 2 \arrow[r] & 3 \end{tikzcd} .$$

It's all the same, as these both admit right anodyne maps into

$$\begin{tikzcd} 0 \arrow[r] \arrow[d] \arrow[dr, bend left] & 1 \arrow[d] \\ 2 \arrow[r] \arrow[ur, Rightarrow] & 3 \arrow[ul, Rightarrow] \end{tikzcd} .$$

Now, fix a map $p(012)\colon (2 \leftarrow 0 \rightarrow 1) \rightarrow \mathcal{C}$. Let $p(123)\colon (2 \rightarrow 3 \leftarrow 1) \rightarrow \mathcal{C}$ be a colimit of $p(12)$. There arises a map $p(0123)$ from the square mentioned above; let $p(01234)$ be a colimit of this. For sake of visualiation, the domain of $p(01234)$ is the simplcial set

$$\begin{tikzcd} 0 \arrow[rr] \arrow[dr] \arrow[dd] & & 1 \arrow[dl] \arrow[dd] \\ & 4 & \\ 2 \arrow[rr] \arrow[ur] & & 3 \arrow[ul] \end{tikzcd}$$

with all triangles filled in. Now, consider the diagram

$$\begin{tikzcd} \mathcal{C}_{p(01234)/} \arrow[r, "\mathrm{TFib}"] \arrow[d, "\mathrm{TFib}"] & \mathcal{C}_{p(0123)/} \arrow[r] \arrow[d] & \mathcal{C}_{p(123)/} \arrow[d, "\mathrm{TFib}"] \\ \mathcal{C}_{p(0124)/} \arrow[r] & \mathcal{C}_{p(012)/} \arrow[r] & \mathcal{C}_{p(12)/} \end{tikzcd} .$$

The left vertical map is a trivial fibration as it is the restriction along the right anodyne map

$$\begin{tikzcd} 0 \arrow[rr] \arrow[dr] \arrow[dd] & & 1 \arrow[dl] \\ & 4 & \\ 2 \arrow[ur] & & \end{tikzcd} \subset \begin{tikzcd} 0 \arrow[rr] \arrow[dr] \arrow[dd] & & 1 \arrow[dl] \arrow[dd] \\ & 4 & \\ 2 \arrow[rr] \arrow[ur] & & 3 \arrow[ul] \end{tikzcd} .$$

The other indicated maps are trivial fibrations by choice. Now, the right square is a pullback, so the middle vertical map is a trivial fibration. It follows by 2-out-of-3 that $\mathcal{C}_{p(0124)/} \rightarrow \mathcal{C}_{p(012)/}$ is a trivial fibration, so that $p(0124) = \operatorname{colim} p(012)$. $\square$

The Grothendieck Construction and Colimits

It turns out that unstraightening is related to taking colimits of spaces. But before we discuss this we review the case for ordinary categories.

If $F\colon \mathcal{C}^{\mathrm{op}} \to \mathrm{Set}$ is some functor, we might ask what $\operatorname{colim} F$ is. Actually, there is a "formula" for this. Namely,

$$\operatorname{colim} F = \left(\coprod_{c \in \mathcal{C}} Fc \right) /\!\sim,$$

where $\sim$ is the equivalence relation generated by forcing $x \sim x'$ whenever $x \in Fc$, $x' \in Fc'$, and there is $f\colon c \to c'$ such that $F(f)(x') = x$.

Now, let $\int\colon \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathrm{Set}) \to \mathrm{Cat}_{/\mathcal{C}}$ be the Grothendieck construction. Then $\int F$ has as objects the pairs $\langle c \in \mathcal{C}, x \in F(c) \rangle$, with a morphism $\langle c, x \rangle \to \langle c', x' \rangle$ being $f\colon c \to c'$ such that $F(f)(x') = x$. As a result, the following is immediate.

Proposition 3.1. A choice for $\operatorname{colim}\colon \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathrm{Set}) \to \mathrm{Set}$ is given by

$$\mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathrm{Set}) \xrightarrow{\int} \mathrm{Cat}_{/\mathcal{C}} \xrightarrow{\mathrm{proj}} \mathrm{Cat} \xrightarrow{\pi_0} \mathrm{Set}\ .$$

□

Unstraightening and Colimits

As should be no surprise by now, this generalizes to colimits of functors $F\colon \mathcal{C}^{\mathrm{op}} \to \mathrm{Spaces}$, where $\mathcal{C}$ is a quasicategory. However, there is a slight subtlety. What's true is the following.

Fact 4.1. Let $F\colon \mathcal{C}^{\mathrm{op}} \to \mathrm{Spaces}$. Then a choice for $\operatorname{colim} F$ is given by the space represented by $\operatorname{Un} F$.

The subtlety comes from the fact that $\operatorname{Un} F$ is generally not going to be a Kan complex, so if we write "$\operatorname{colim} F = \operatorname{Un} F$", we must be careful where this "=" is taking place.

For more on this you can take a look at the detailed work on Grothendieck construction and colimits in [GHN15] or [Ma15].

Homotopy colimits

One of the motivations mentioned for introducing higher category theory is the fact that many categorical constructions are not homotopically well behaved. For instance, $\operatorname{colim}\colon \mathrm{Fun}(I, \mathrm{Top}) \to \mathrm{Top}$ does not preserve weak (or homotopy) equivalences. The classical way of dealing with this was to use homotopy co/limits.

Example 5.1. Consider the diagram $\mathrm{pt} \leftarrow S^1 \xrightarrow{2} S^1$. The resulting pushout is just pt. Morally, the homotopy pushout would be a homotopy kernel of $S^1 \xrightarrow{2} S^1$. Of course, this is just given by the map $S^1 \to \mathbb{RP}^2$.

Homotopy colimits can be defined for any suitable category with weak equivalences, although they are commonly defined in the context of model categories. Since we think of model categories as being models of $(\infty, 1)$-categories, what we want to be true is something like the following.

Dream Theorem 5.2. *To every model category $\mathcal{M}$, there is associated an $(\infty, 1)$-category $\mathcal{M}_\infty$, such that*

- $\mathrm{ho}(\mathcal{M}_\infty) = \mathrm{ho}(\mathcal{M})$,
- *For any functor $F \colon I \to \mathcal{M}_\infty$, we have*

$$\mathrm{ho}(\operatorname{colim} F) = \operatorname{hocolim}(\mathrm{ho}\, F).$$

We will describe a way to associate to a model category a quasicategory, and assert the above to be true, so the first part is definitely true. But the second one seems a little questionable.

The quasicategory associated to a Model Category

There are several ways to take a model structure and build a quasi-category out of it. There an approach in Section 8 [Re01], which is not the best approach as it requires the model structure to be simplicial and relies on Reedy fibrant replacements.

The approach we follow here is able do avoid that by relying on *double subdivision* and was introduced by [BK12]. There is also a nice discussion in [Me16].

Recall that for a category $\mathcal{C}$, we have defined $\mathcal{NC}$ to be the bisimplicial set

$$(\mathcal{NC})_{p,q} = N((\mathcal{C}^{[p]})^{\mathrm{core}})_q.$$

This was given as a motivating example for introducing Segal spaces and complete Segal spaces. Taking the core restricts certain maps to be isomorphisms; an obvious generalization would then be to restrict certain maps to be only weak equivalences, suitably defined. This doesn't quite give the answer we want, but it's reasonably close.

Definition 6.1. A *relative category* is a pair $(\mathcal{C}, \mathcal{W})$, where $\mathcal{C}$ is a category and $\mathcal{W} \subset \mathcal{C}$ is a subcategory containing all objects.

There arises a category RelCat of relative categories and weak equivalence preserving functors. It turns out that relative categories can be used to model

all $(\infty, 1)$-categories, but describing this would take us too far afield. What we can say is the following.

Fact 6.2. There is a functor $\mathcal{N}_\xi \colon \mathrm{RelCat} \to s\mathcal{S}$ such that if $(\mathcal{M}, \mathcal{W})$ is a model category, then $\mathcal{N}_\xi(\mathcal{M}, \mathcal{W})$ is a complete Segal space. In particular, taking the first row of $\mathcal{N}_\xi(\mathcal{M}, \mathcal{W})$ gives a quasicategory. Moreover, $\mathrm{ho}(\mathcal{N}_\xi(\mathcal{M}, \mathcal{W})) = \mathrm{ho}(\mathcal{M}, \mathcal{W})$.

This is a correct construction of a quasicategory from a model category. It agrees with other methods of obtaining an $(\infty, 1)$-category from a model category, such as hammock localization.

Now, we briefly describe the functor $\mathcal{N}_\xi$. It is not quite the obvious generalization of $\mathcal{N}$, but is very similar.

Definition 6.3. Let (P, I) be a relative poset. the *terminal*, resp., *initial*, subdivision $\xi_t P$, resp., $\xi_i P$, is a relative poset with

- Objects the functors $[n] \to P$.
- Morphisms $f \to g$ being

$$\begin{tikzcd} {[n]} \arrow[rr] \arrow[dr, "f"'] & & {[m]} \arrow[dl, "g"] \\ & P & \end{tikzcd},$$

  resp.,

$$\begin{tikzcd} {[m]} \arrow[rr] \arrow[dr, "g"'] & & {[n]} \arrow[dl, "f"] \\ & P & \end{tikzcd}.$$

- Weak equivalences the morphisms such that $f(n) \to g(m)$, resp., $g(0) \to f(0)$, is in I.

The *double subdivision* of (P, I) is $\xi(P, I) = \xi_t \xi_i (P, I)$.

Now, consider $[p] \times [q]$ as a relative poset where the "vertical" maps are weak equivalences; we write this as $([p] \times [q], [q])$, although this isn't very precise. Notice that

$$\mathcal{N}(\mathcal{C})_{p,q} = \mathrm{RelCat}(([p] \times [q], [q]), (\mathcal{C}, \mathcal{C}^{\mathrm{core}})).$$

Now, the appropriate generalization of $\mathbb{N}$ turns out to be the following.

Definition 6.4. Define $\mathcal{N}_\xi \colon \mathrm{RelCat} \to \mathcal{S}\mathrm{sSet}$ by

$$\mathcal{N}_\xi(\mathcal{C}, \mathcal{W})_{p,q} = \mathrm{RelCat}(\xi([p] \times [q], [q]), (\mathcal{C}, \mathcal{W})).$$

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